Calcium chloride: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Calcium chloride: Patient drug information" and "Calcium chloride: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: Canada

- Calciject

Pharmacologic Category

- Calcium Salt;
- Electrolyte Supplement, Parenteral

Dosing: Adult

Note: One gram of calcium chloride salt is equal to 270 mg of elemental calcium.

Dosages are expressed in terms of the calcium chloride salt (unless otherwise specified as elemental calcium). Dosages expressed in terms of the calcium chloride salt are based on a solution concentration of 100 mg/mL (10%) containing 1.4 mEq (27 mg) elemental calcium per mL.

Beta-blocker toxicity

Beta-blocker toxicity (off-label use): Based on limited data: **IV:** Initial: Using a 10% solution: 1 to 2 **g** (10 to 20 **mL** of a 10% solution) or 20 **mg/kg** (0.2 **mL/kg** of a 10% solution) over 5 to 10 minutes; may repeat every 10 to 20 minutes for 3 to 4 additional doses **or** initiate a continuous infusion of 20 to 40 **mg/kg/hour** titrated to improve hemodynamic response (Ref).

Calcium channel blocker toxicity

Calcium channel blocker toxicity (off-label use): Based on limited data: **IV:** Initial: Using a 10% solution: 1 to 2 **g** (10 to 20 **mL** of a 10% solution) or 20 **mg/kg** (0.2 **mL/kg** of a 10% solution) over 5 to 10 minutes; may repeat every 10 to 20 minutes for 3 to 4 additional doses **or** initiate a continuous infusion of 20 to 40 **mg/kg/hour** titrated to improve hemodynamic response (Ref). **Note:** Maintain serum ionized calcium at 1.5 to 2 times the upper limit of normal (Ref).

Calcium replacement, during CRRT with citrate-based regional anticoagulation

Calcium replacement, during CRRT with citrate-based regional anticoagulation (off-label dose):

Intracircuit, posthemodialysis filter (returning to the patient) or IV through a separate central line: Prior to the initiation of CRRT, ensure adequate systemic ionized calcium concentrations (drawn directly from the patient). During CRRT, use a calcium chloride continuous infusion sliding scale to maintain systemic ionized calcium between ~3.6 to 5.2 **mg/dL** (~0.9 to 1.3 mmol/L); monitor systemic ionized calcium every 6 hours (or more frequently when indicated) (Ref). Refer to institutional protocols.

Hydrofluoric acid exposure

Hydrofluoric acid exposure (off-label use): Note: Consultation with a clinical toxicology or poison control center is recommended prior to the use of calcium chloride for hydrofluoric acid exposure.

Systemic toxicity: **Note:** Calcium chloride has been used in the treatment of systemic toxicity secondary to hydrofluoric acid exposure (Ref); however, calcium gluconate may be preferred due to the potential for more severe extravasation with calcium chloride.

IV: Exact dose has not been established; clinicians should tailor dose based on patient-specific needs (Ref). Bolus doses of up to 4 g have been required (Ref); repeat as needed based on symptoms of toxicity (eg, cardiac arrhythmias) and serum calcium concentration.

Hydrofluoric acid burns (severe): **Intra-arterial:** 10% solution: Add 10 **mL** of a 10% solution in 40 to 50 mL of D_5W; infuse over 4 hours into the artery that provides the vascular supply to the affected area. Pain usually resolves by the end of the infusion; repeat if pain recurs (Ref); calcium gluconate may be preferred due to the potential for vessel injury and extravasation (Ref). **This intervention should be used only by those accustomed to this technique. Care should be taken to avoid the extravasation.** A poison information center or clinical toxicologist should be consulted prior to implementation.

Hyperkalemia, severe/emergent

Hyperkalemia , severe/emergent (off-label use): Note: Use in patients with hyperkalemia-associated ECG changes **or** serum potassium >6.5 mEq/L (Ref). Stabilizes myocardial cell membrane without impacting plasma potassium concentrations; must combine with insulin/dextrose to decrease plasma potassium levels and other therapies to eliminate potassium from body (Ref). Perform continuous cardiac monitoring and obtain serial ECGs (Ref).

IV, Intraosseous: Initial: 0.5 to 1 g over 2 to 5 minutes; may repeat after 5 minutes if ECG changes persist or recur, then every 30 to 60 minutes as needed (Ref).

Hypocalcemia, acute

Hypocalcemia, acute (alternative agent):

Note: For use in patients with severe symptoms of hypocalcemia (eg, tetany, seizures, carpopedal spasm), ECG abnormalities (eg, QTc prolongation, arrhythmia), or an acute decrease in albumin-corrected serum calcium levels to <7 to 7.5 mg/dL (<1.75 to 1.87 mmol/L) when serious complications may occur if untreated (eg, following neck surgery). In patients **without** a central line, calcium gluconate is preferred over calcium chloride due to the potential for more severe extravasation with calcium chloride. Correct concurrent hypomagnesemia if present (Ref). **Do not** use IV calcium as initial therapy in patients with chronic kidney disease who are asymptomatic or who have stable hypocalcemia with only mild symptoms (eg, paresthesias) (Ref).

Initial bolus dose(s): **IV:** Add 1 g (10 **mL** of a 10% solution) to 100 mL of D5W or NS (equivalent to 270 mg **elemental calcium**). Infuse via a central or large vein over 10 to 20 minutes; may repeat bolus dose after 10 to 60 minutes if symptoms persist (Ref). If hypocalcemia is expected to persist (eg, hypoparathyroidism, pancreatitis), follow bolus dose(s) with a continuous IV calcium infusion (Ref).

Continuous infusion: IV**:** Add 4 g (40 **mL** of a 10% solution) to 960 mL of D5W or NS (equivalent to ~1 g **elemental calcium** in a final total volume of 1,000 mL). Initiate infusion at 50 to 100 mL/hour (equivalent to ~50 to 100 mg/hour of **elemental calcium**) via a central or large vein; adjust dose to maintain albumin-corrected serum calcium levels at the low end of normal (Ref). Initiate oral calcium and vitamin D supplements as soon as possible; once an effective regimen is achieved, taper IV calcium infusion slowly (eg, over 24 to 48 hours) (Ref). **Note:** Instructions for preparing calcium chloride infusion are adapted from recommendations for calcium gluconate (Ref).

Dosing: Kidney Impairment: Adult

Note: **Do not** use IV calcium as initial therapy in patients with chronic kidney disease who are asymptomatic or who have stable hypocalcemia with only mild symptoms (eg, paresthesias) (Ref).

Initiate with lowest dose of the recommended dosage range.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. No initial dosage adjustment necessary; subsequent doses should be guided by serum calcium concentrations.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Calcium chloride: Pediatric drug information")

Dosage guidance:

Dosing: 1,000 mg of calcium chloride = 270 mg of **elemental** calcium = 14 mEq of **elemental** calcium.

Safety: Dosing may be presented as calcium chloride or as **elemental** calcium and expressed as mg or mEq; use caution.

Calcium channel blocker toxicity

Calcium channel blocker toxicity: Infants, Children, and Adolescents: Dose expressed as **calcium chloride**: IV: 10 to 20 mg/kg/dose infused over 5 to 10 minutes, maximum dose: 2,000 mg/dose; if effective, may repeat bolus doses every 10 to 15 minutes or initiate continuous IV infusion at 20 to 50 mg/kg/**hour** (Ref).

Cardiac arrest in the presence of hyperkalemia or hypocalcemia, hypermagnesemia, or calcium channel blocker toxicity

Cardiac arrest in the presence of hyperkalemia or hypocalcemia, hypermagnesemia, or calcium channel blocker toxicity (PALS recommendations):

Note: Calcium chloride is the preferred salt in urgent situations; not recommended in the absence of hyperkalemia, hypocalcemia, hypermagnesemia, or calcium channel blocker toxicity due to the lack of improved survival (Ref).

Infants, Children, and Adolescents: Dose expressed as **calcium chloride**: IV, Intraosseous: 20 mg/kg/dose, maximum dose: 2,000 mg/dose; may consider repeat dose in 10 minutes if clinically necessary (Ref).

Hypocalcemia, symptomatic

Hypocalcemia, symptomatic:

Note: Calcium gluconate may be the preferred salt in non-cardiac arrest settings due to extravasation risks which are greater during peripheral venous administration and due to potential for metabolic acidosis (Ref).

Infants, Children, and Adolescents: Dose expressed as **calcium chloride**: IV: 10 to 20 mg/kg/dose, maximum dose: 1,000 mg/dose; repeat as needed based on clinical response and desired effect; some experts suggest every 4 to 6 hours as needed (Ref).

Low cardiac output

Low cardiac output (adjunct therapy): Limited data available: Infants, Children, and Adolescents: Dose expressed as **calcium chloride**: Continuous IV infusion: 5 to 10 mg/kg/**hour**. Dosing based on retrospective report of pediatric calcium infusions (n=116 [51 infusions in ages 1 month to 17 years]) for low cardiac output of surgical and nonsurgical etiologies; study results did not show a statistically significant difference in improvement of cardiac output (Ref).

Parenteral nutrition, maintenance calcium requirement

Parenteral nutrition (PN), maintenance calcium requirement:

Note: Calcium gluconate is the preferred salt formulation for the preparation of PN since it is less reactive than calcium chloride (at equivalent amounts) and allows for improved solubility with phosphate salts (Ref).

Infants and Children ≤50 kg: Dose expressed as **elemental calcium**: IV: 0.5 to 4 **mEq**/kg/**day** as an additive to PN solution (Ref).

Children >50 kg and Adolescents: Dose expressed as **elemental calcium**: IV: 10 to 20 **mEq/day** as an additive to PN solution (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage recommendations in the manufacturer’s labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage recommendations in the manufacturer’s labeling.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Postmarketing:

Cardiovascular: Cardiac arrhythmia (Carlon 1978), hypotension (Carlon 1978), peripheral vasodilation

Dermatologic: Cutaneous calcification (Ehsani 2006)

Gastrointestinal: Medicine-like taste (calcium taste)

Local: Injection-site reaction, localized burning

Contraindications

Patients with ventricular fibrillation; patients with asystole and electromechanical dissociation; concomitant use of IV calcium chloride with ceftriaxone in neonates (≤28 days of age).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• Extravasation: Vesicant; ensure proper catheter or needle position prior to and during infusion. Avoid extravasation. Extravasation may result in severe necrosis and sloughing. Monitor the IV site closely.

Disease-related concerns:

• Acidosis: Use with caution in patients with respiratory acidosis, renal impairment, or respiratory failure; acidifying effect of calcium chloride may potentiate acidosis.

• Hydrofluoric acid burns: Calcium chloride can be administered intra-arterially for severe hydrofluoric acid exposures. However, the chloride salt should never be injected into tissues (do **not** inject SUBQ or IM) due to the risk of tissue necrosis (Smith 2019). Consultation with a clinical toxicologist or a poison information center before using calcium chloride to treat hydrofluoric acid toxicity is recommended.

• Hyperphosphatemia: Use with caution in patients with severe hyperphosphatemia as elevated levels of phosphorus and calcium may result in soft tissue and pulmonary arterial calcium-phosphate precipitation.

• Hypokalemia: Use with caution in patients with severe hypokalemia as acute rises in serum calcium levels may result in life-threatening cardiac arrhythmias.

• Hypomagnesemia: Hypomagnesemia is a common cause of hypocalcemia; therefore, correction of hypocalcemia may be difficult in patients with concomitant hypomagnesemia. Evaluate serum magnesium and correct hypomagnesemia (if necessary), particularly if initial treatment of hypocalcemia is refractory.

• Renal impairment: Use with caution in patients with chronic renal failure to avoid hypercalcemia; frequent monitoring of serum calcium and phosphorus is necessary.

Concurrent drug therapy issues:

• Ceftriaxone: Ceftriaxone may complex with calcium causing precipitation. Fatal lung and kidney damage associated with calcium-ceftriaxone precipitates has been observed in premature and term neonates. Due to reports of precipitation reaction in neonates, do not coadminister ceftriaxone with calcium-containing solutions, even via separate infusion lines/sites or at different times in any neonate. Ceftriaxone should not be administered simultaneously with any calcium-containing solution via a Y-site in any patient. However, ceftriaxone and calcium-containing solutions may be administered sequentially of one another for use in patients **other than neonates** if infusion lines are thoroughly flushed (with a compatible fluid) between infusions.

• Digoxin: Use with caution in digitalized patients; hypercalcemia may precipitate cardiac arrhythmias.

Dosage form specific issues:

• Aluminum: The parenteral product may contain aluminum; toxic aluminum concentrations may be seen with high doses, prolonged use, or renal dysfunction. Premature neonates are at higher risk due to immature renal function and aluminum intake from other parenteral sources. Parenteral aluminum exposure of >4 to 5 mcg/kg/day is associated with CNS and bone toxicity; tissue loading may occur at lower doses (Federal Register, 2002). See manufacturer's labeling.

Other warnings/precautions:

• Appropriate product selection: Multiple salt forms of calcium exist; close attention must be paid to the salt form when ordering and administering calcium; incorrect selection or substitution of one salt for another without proper dosage adjustment may result in serious over or under dosing.

• Duration of use: Avoid metabolic acidosis (ie, administer only up to 2 to 3 days then change to another calcium salt).

• IV administration: For IV use only; do not inject SUBQ or IM. Avoid too rapid IV administration (do not exceed 100 mg/minute except in emergency situations); arrythmia, bradycardia, hypotension, syncope, and vasodilation may occur.

Dosage Forms Considerations

1 g calcium chloride = elemental calcium 270 mg = calcium 14 mEq = calcium 6.8 mmol

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 10% (10 mL)

Solution, Intravenous [preservative free]:

Generic: 10% (10 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Calcium Chloride Intravenous)

10% (per mL): $0.90 - $2.43

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Calciject: 10% (10 mL, 50 mL)

Generic: 10% (10 mL)

Administration: Adult

IV: For IV administration only. Not for IM or SUBQ administration (severe necrosis and sloughing may occur). Avoid rapid administration (do not exceed 100 mg/minute except in emergency situations). For intermittent IV infusion, infuse diluted solution over 1 hour or no greater than 45 to 90 mg/kg/hour (0.6 to 1.2 mEq/kg/hour); administration via a central or deep vein is preferred; do not use small hand or foot veins for IV administration (severe necrosis and sloughing may occur). Typical rates of administration may vary with indication; refer to institutional protocol. Monitor ECG if calcium is infused faster than 2.5 mEq/minute; **stop the infusion if the patient complains of pain or discomfort.** Warm solution to body temperature prior to administration. **Do not infuse calcium chloride in the same IV line as phosphate-containing solutions.**

Hydrofluoric acid burns (severe) (off-label use/route): Intra-arterial: Requires radiology to place an arterial catheter in an artery supplying blood to the area of exposure; infuse over 4 hours (Ref). **This intervention should be used only by those trained in this technique. Care should be taken to avoid the extravasation.** A poison information center or clinical toxicologist should be consulted prior to implementation.

Vesicant; ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation.

Extravasation management:

Early/acute calcium extravasation: If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote (Ref).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 to 1.7 mL (15 units/mL) as 5 separate 0.2 to 0.3 mL injections (using a tuberculin syringe) around the site of extravasation; if IV catheter remains in place, administer intravenously through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Delayed calcium extravasation: Closely monitor site; most calcifications spontaneously resolve. However, if a severe manifestation of calcinosis cutis occurs, may initiate sodium thiosulfate antidote.

Sodium thiosulfate: IV: 12.5 g over 30 minutes, administered 3 times per week; may increase gradually to 25 g 3 times per week; monitor for non-anion gap acidosis, hypocalcemia, severe nausea (Ref).

Preparation for Administration: Adult

IV: For intermittent IV infusion, dilute in NS or D5W to a concentration of 10 mg/mL (Ref).

Administration: Pediatric

Parenteral: Do not use scalp vein or small hand or foot veins for IV administration; central-line administration is the preferred route. Not for endotracheal administration. Do not inject calcium salts IM or administer SUBQ since severe necrosis and sloughing may occur; extravasation of calcium can result in severe necrosis and tissue sloughing. Stop the infusion if the patient complains of pain or discomfort. Warm solution to body temperature prior to administration. Do not infuse calcium chloride in the same IV line as phosphate-containing solutions or with ceftriaxone.

IV: Avoid rapid administration; do not exceed 100 mg/minute except in emergency situations. For patients in cardiac arrest or with calcium channel blocker toxicity, administer as a bolus over 5 to 10 minutes via a central line or intraosseous route (Ref). For intermittent infusions, administer diluted solutions over >30 to 60 minutes via a central line (Ref).

Continuous IV infusion: Administer as a continuous infusion via an infusion pump.

Parenteral nutrition solution: Calcium chloride is not routinely used in the preparation of parenteral nutrition. Calcium-phosphate stability and precipitation in parenteral nutrition solution are dependent upon the pH of the solution, temperature, relative concentration of each ion, and salt formulation for Ca and PO_4. Calcium chloride has a higher propensity to precipitate than calcium gluconate. The pH of the solution is primarily dependent upon the amino acid concentration and composition. The higher percentages of amino acids result in solutions with a lower pH, which decreases the risk for calcium and phosphate precipitation. Individual commercially available amino acid solutions vary significantly with respect to pH lowering potential and consequent calcium phosphate compatibility; consult product-specific labeling and solubility curve data for additional information (Ref).

Vesicant; ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation.

Early/acute calcium extravasation: If acute extravasation occurs, stop infusion immediately and disconnect (leave needle/cannula in place); gently aspirate extravasated solution (do **NOT** flush the line); initiate hyaluronidase antidote; remove needle/cannula; apply dry warm compresses; elevate extremity (Ref).

Delayed calcium extravasation: If delayed extravasation suspected, closely monitor site; most calcifications spontaneously resolve. However, if a severe manifestation of cutaneous calcinosis occurs, may initiate sodium thiosulfate antidote (Ref).

Preparation for Administration: Pediatric

Parenteral: IV infusion: Dilute to a maximum concentration of 20 mg/mL.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Store intact vials at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Do not refrigerate solutions; IV infusion solutions in D5W, LR, NS, or other appropriate solutions are stable for 24 hours at room temperature.

Although calcium chloride is not routinely used in the preparation of parenteral nutrition, it is important to note that phosphate salts may precipitate when mixed with calcium salts. Solubility is improved in amino acid parenteral nutrition solutions. Check with a pharmacist to determine compatibility.

Pharmacy supply of emergency antidotes: Guidelines suggest that at least 10 g of calcium chloride be stocked. Suggested amount is stated to be a sufficient quantity to provide antidotal treatment for 1 patient weighing 100 kg for an initial 8- to 24-hour period (Dart 2018); actual amount to be stocked should take into account site-specific and population-specific needs.

Use: Labeled Indications

Hypocalcemia, acute: Treatment of acute symptomatic hypocalcemia.

Use: Off-Label: Adult

Beta-blocker toxicity; Calcium channel blocker toxicity; Calcium replacement, during CRRT with citrate-based regional anticoagulation; Hydrofluoric acid exposure; Hyperkalemia, severe/emergent

Medication Safety Issues

Sound-alike/look-alike issues:

Calcium chloride may be confused with calcium gluconate

Administration issues:

Calcium chloride may be confused with calcium gluconate.

Confusion with the different intravenous salt forms of calcium has occurred. There is a threefold difference in the primary cation concentration between **calcium chloride** (in which 1 g = 14 mEq [270 mg] of elemental Ca++) and **calcium gluconate** (in which 1 g = 4.65 mEq [93 mg] of elemental Ca++).

Prescribers should specify which salt form is desired. To prevent medication errors, dosages should be expressed either as mg or grams of the salt form for most indications. Dosages expressed as mEq may be used for nutrition.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Baloxavir Marboxil: Polyvalent Cation Containing Products may decrease serum concentrations of Baloxavir Marboxil. *Risk X: Avoid*

Bisphosphonate Derivatives: Polyvalent Cation Containing Products may decrease serum concentrations of Bisphosphonate Derivatives. Management: Avoid administration of oral medications containing polyvalent cations within: 2 hours before or after tiludronate/clodronate/etidronate; 60 minutes after oral ibandronate; or 30 minutes after alendronate/risedronate. *Risk D: Consider Therapy Modification*

Cabotegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Cabotegravir. Management: Administer polyvalent cation containing products at least 2 hours before or 4 hours after oral cabotegravir. *Risk D: Consider Therapy Modification*

Calcium Acetate: Calcium Salts may increase adverse/toxic effects of Calcium Acetate. *Risk X: Avoid*

Calcium Channel Blockers: Calcium Salts may decrease therapeutic effects of Calcium Channel Blockers. *Risk C: Monitor*

Calcium Polystyrene Sulfonate: Polyvalent Cation Containing Products may decrease therapeutic effects of Calcium Polystyrene Sulfonate. *Risk C: Monitor*

Cardiac Glycosides: Calcium Salts may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

CefTRIAXone: Calcium Salts (Intravenous) may increase adverse/toxic effects of CefTRIAXone. Ceftriaxone binds to calcium forming an insoluble precipitate. Management: Use of ceftriaxone is contraindicated in neonates (28 days of age or younger) who require (or are expected to require) treatment with IV calcium-containing solutions. In older patients, flush lines with compatible fluid between administration. *Risk X: Avoid*

Deferiprone: Polyvalent Cation Containing Products may decrease serum concentrations of Deferiprone. Management: Separate administration of deferiprone and oral medications or supplements that contain polyvalent cations by at least 4 hours. *Risk D: Consider Therapy Modification*

DOBUTamine: Calcium Salts may decrease therapeutic effects of DOBUTamine. *Risk C: Monitor*

Eltrombopag: Polyvalent Cation Containing Products may decrease serum concentrations of Eltrombopag. Management: Administer eltrombopag at least 2 hours before or 4 hours after oral administration of any polyvalent cation containing product. *Risk D: Consider Therapy Modification*

Elvitegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Elvitegravir. Management: Administer elvitegravir-containing products 2 hours before, or 2 to 6 hours after, the administration of polyvalent cation containing products. *Risk D: Consider Therapy Modification*

Multivitamins/Fluoride (with ADE): May increase serum concentrations of Calcium Salts. Calcium Salts may decrease serum concentrations of Multivitamins/Fluoride (with ADE). More specifically, calcium salts may impair the absorption of fluoride. Management: Avoid eating or drinking dairy products or consuming vitamins or supplements with calcium salts one hour before or after of the administration of fluoride. *Risk D: Consider Therapy Modification*

Multivitamins/Minerals (with ADEK, Folate, Iron): May increase serum concentrations of Calcium Salts. *Risk C: Monitor*

PenicillAMINE: Polyvalent Cation Containing Products may decrease serum concentrations of PenicillAMINE. Management: Separate the administration of penicillamine and oral polyvalent cation containing products by at least 1 hour. *Risk D: Consider Therapy Modification*

Raltegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Raltegravir. Management: Administer raltegravir 2 hours before or 6 hours after administration of the polyvalent cations. Dose separation may not adequately minimize the significance of this interaction. *Risk D: Consider Therapy Modification*

Roxadustat: Polyvalent Cation Containing Products may decrease serum concentrations of Roxadustat. Management: Administer roxadustat at least 1 hour after the administration of oral polyvalent cation containing products. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Medronate: Coadministration of Calcium Salts and Technetium Tc 99m Medronate may alter diagnostic results. Management: Consider performing imaging with technetium Tc 99m medronate after cation levels have normalized (eg, 3 to 5 half-lives of the cation) in patients with high calcium levels caused by coadministration of calcium salts. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Oxidronate: Coadministration of Calcium Salts and Technetium Tc 99m Oxidronate may alter diagnostic results. Management: Consider performing imaging with technetium Tc 99m oxidronate after cation levels have normalized (eg, 3 to 5 half-lives of the cation) in patients with high calcium levels caused by coadministration of calcium salts. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Pyrophosphate: Coadministration of Calcium Salts and Technetium Tc 99m Pyrophosphate may alter diagnostic results. Management: Consider performing imaging with technetium Tc 99m pyrophosphate after cation levels have normalized (eg, 3 to 5 half-lives of the cation) in patients with high calcium levels caused by coadministration of calcium salts. *Risk D: Consider Therapy Modification*

Thiazide and Thiazide-Like Diuretics: May increase serum concentrations of Calcium Salts. *Risk C: Monitor*

Trientine: Polyvalent Cation Containing Products may decrease serum concentrations of Trientine. Management: Avoid concomitant use of trientine and polyvalent cations. If oral iron supplements are required, separate the administration by 2 hours. For other oral polyvalent cations, give trientine 1 hour before, or 1 to 2 hours after the polyvalent cation. *Risk D: Consider Therapy Modification*

Unithiol: May decrease therapeutic effects of Polyvalent Cation Containing Products. *Risk X: Avoid*

Vitamin D Analogs: Calcium Salts may increase adverse/toxic effects of Vitamin D Analogs. *Risk C: Monitor*

Pregnancy Considerations

Calcium crosses the placenta. The amount of calcium reaching the fetus is determined by maternal physiological changes. Calcium requirements are the same in pregnant and nonpregnant females (IOM 2011).

Information related to use as an antidote in pregnancy is limited. In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant women if there is a clear indication for use and should not be withheld because of fears of teratogenicity (Bailey 2003).

Breastfeeding Considerations

Calcium is excreted in breast milk. The amount of calcium in breast milk is homeostatically regulated and not altered by maternal calcium intake. Calcium requirements are the same in lactating and nonlactating females (IOM 2011).

Monitoring Parameters

Serum calcium and ionized calcium; serum calcium every 4 hours in patients with renal impairment; albumin; serum phosphate; magnesium (to facilitate calcium repletion); ECG when appropriate. Monitor infusion site.

Calcium channel blocker toxicity, beta-blocker toxicity (off-label uses): Monitor hemodynamic response; monitor serum ionized calcium levels every 30 minutes initially then every 2 hours and maintain ionized calcium ~1.5 to 2 times the ULN; avoid severe hypercalcemia (ionized calcium levels >2 times ULN) (AHA [Lavonas 2023]; Kerns 2007).

Reference Range

Serum total calcium: 8.6 to 10.2 mg/dL (SI: 2.2 to 2.6 mmol/L). **Note:** Due to a poor correlation between the serum ionized calcium (free) and total serum calcium, particularly in states of low albumin or acid/base imbalances, direct measurement of ionized calcium is recommended.

Serum ionized calcium: 4.48 to 4.92 mg/dL (SI: 1.12 to 1.23 mmol/L).

In low albumin states, the corrected **total** serum calcium may be estimated by the following equation (assuming a normal albumin of 4 g/dL [SI: 40 g/L]).

Corrected total calcium (mg/dL) = measured total calcium (mg/mL) + 0.8 [4 - measured serum albumin (g/dL)].

or

Corrected total calcium (mmol/L) = measured total calcium (mmol/L) + 0.02 [40-measured serum albumin (g/L)].

Mechanism of Action

Moderates nerve and muscle performance via action potential excitation threshold regulation.

In hydrofluoric acid (hydrogen fluoride) exposures, calcium chloride provides an exogenous source of calcium which can bind fluoride ions as well as treat and prevent complications secondary to hypocalcemia; intra-arterial administration can reduce the penetration of the fluoride ion into tissues and prevent or reduce tissue destruction and pain (Vance 1986; Wu 2010).

Pharmacokinetics (Adult Data Unless Noted)

Distribution: Primarily in skeleton (99%).

Protein binding: ~40%, primarily to albumin.

Excretion: Primarily feces (80% as insoluble calcium salts); urine (20%) (IOM 2011).

Patient Counseling Points

(For additional information see "Calcium chloride: Patient drug information")

What is this drug used for?

• It is used to treat or prevent low calcium levels.

• It is used to protect the heart from high potassium levels.

• It is used to protect the heart and lungs from high magnesium levels.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Bad taste

• Hot flashes

• Lack of appetite

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• High calcium like weakness, confusion, fatigue, headache, nausea and vomiting, constipation, or bone pain

• Dizziness

• Passing out

• Abnormal heartbeat

• Kidney stone like back pain, abdominal pain, or blood in the urine

• Severe injection site pain, redness, burning, swelling, blisters, skin sores, or leaking of fluid

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Cloruro de calcio | Cloruro de calcio fada | Cloruro de calcio lavimar;
- (AT) Austria: Calciumchlorid fresenius kabi aus;
- (BE) Belgium: Calciclo;
- (BR) Brazil: Calrecia;
- (CH) Switzerland: Calrecia;
- (CL) Chile: Cloruro de calcio;
- (CZ) Czech Republic: Calcium chloratum;
- (DE) Germany: Calciumchlorid | Calciumchlorid Baxter;
- (DO) Dominican Republic: Cloruro de calcio;
- (EC) Ecuador: Calrecia;
- (EE) Estonia: Calcii chloridum;
- (ES) Spain: Cloruro calcico Braun;
- (FR) France: Caddera | Calcium cooper | Chlorure de calacium cooper | Chlorure de calcium aguettant;
- (GR) Greece: Calrecia;
- (HR) Croatia: Calrecia;
- (HU) Hungary: Calcium chloratum;
- (IN) India: Nelcium;
- (IT) Italy: Calcio Cloruro | Calcio cloruro monico;
- (JP) Japan: Calcium chloride kobayashi | Calcium chloride merck hoei | Calcium chloride np | Chlocalin | Encal hikari | Encal otsuka | Otsuka calcium chloride;
- (KR) Korea, Republic of: Huons Calcium Chloride | Jw 3% calcium chloride;
- (LT) Lithuania: Calcium chloratum | Calcium chloridum | Calciumchlorid;
- (LV) Latvia: Calcii chloridum | Calcium chloratum;
- (NO) Norway: Calciumklorid;
- (RU) Russian Federation: Calcium chloratum;
- (SE) Sweden: Calrecia | Kalciumklorid APL;
- (SK) Slovakia: Calcium chloratum;
- (UA) Ukraine: Calcium chloratum;
- (UY) Uruguay: Cloruro de calcio;
- (VN) Viet Nam: Calci clorid

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